

A

## Bounded shear-mixing imbalance diagnostic

Scalar SMII is an enrichment diagnostic; the BLH-shear state-space benchmark remains primary.

### Simple SMII

$$\begin{aligned} & z(\log(1 + S)) \\ & - z(\log(1 + BLH)) \end{aligned}$$

### Constrained SMII

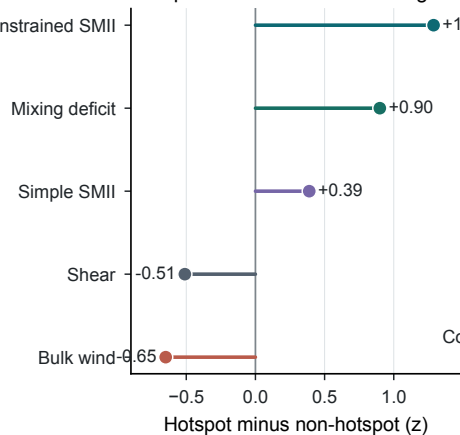
$$\begin{aligned} & -z(\log(1 + BLH)) \\ & + z(\log(1 + S / BLH)) \end{aligned}$$

### Primary test

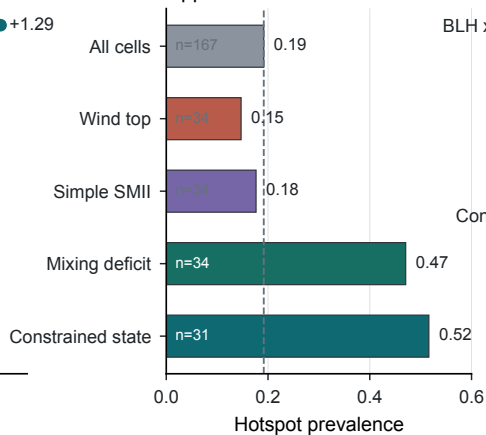
BLH-shear phase state  
plus interaction benchmark

Observation unit: city-state-response cell;  $n = 167$ , hotspot rows = 32. All z-scores use the finite cell table.

## B Hotspot cells are shallow-mixing states



## C Upper-state enrichment



## D State-space model remains strongest

